Sources: the adapters

Trunk section: §6.1 Sources.

Talks to: every domain that needs the outside world: Tools and screeners (the scanner), the Card (quotes, short interest, borrow, news), the Market (index and sector quotes), Research data (EDGAR, 13F, Polygon, index bars), Registers (outcome bars), Quant (chart bars), the Journal (account fills, bars), Platform (health).

Built on: DATA-PIPELINE (every call the card pipeline makes, with endpoints and fallbacks), DATA-FEEDS (qp's bar feeds, warm-up, cache limits, borrow check), DATACHECK (judging each source by its content), CANSLIM §3.1 and §10.4 (the free official sources for O'Neil data), ACCOUNTS (the journal's Alpaca calls).

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1. Why adapters

In the old desk each source is called from inside the stage that needs it, in every tool's process: the same ticker's short interest is fetched once per tool that found it; a fallback order exists in one stage and not another; health is known only when someone presses the data-check button. An adapter puts everything about one endpoint family in one place (a provider with several endpoints, such as Yahoo or Alpaca, has one adapter for each), and every domain asks adapters for **facts**, never calling a source directly.

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2. The adapter contract

Each adapter declares:

Item	Meaning
`families`	the fact families it can produce
`request`	endpoint, method, headers, timeout
`auth`	none, a key (named in the secret store), a User-Agent
`limits`	requests per second or per day; batch size; time budget
`freshness`	how long a fact it produced stays fresh, per family
`parse`	turning the answer into fact values, with alignment and sanity checks
`honest null`	what `none` means (the source answered: nothing) and what `unavailable` means (could not ask), each with a reason
`health`	how its content is judged (ok, degraded, down), on a schedule
`fixtures`	recorded answers, including failures, for CI

Every call is logged with its duration and result; each adapter keeps a counter of calls, failures and the last error, shown on the health page.

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3. Every source

Adapter	Endpoint(s)	Auth	Families	Notes and quirks
tv_scanner	`POST scanner.tradingview.com/america/scan?label-product=screener-stock`	none; browser User-Agent, Origin and Referer of tradingview.com; 15 s	`candidates` (per screener), `quote` (32 columns, batches of 200 by symbol), index and sector-ETF quotes	**ignores unknown columns silently** (verify the column list); refuse a batch whose rows differ in length from the columns asked (D-S3); warn when two range windows are identical on most rows; `short_percentage_of_float` never filled; longer-window highs not split-adjusted
tv_news	`news-headlines.tradingview.com/v2/headlines`	none	`news`	—
yahoo_quote	`quoteSummary`	cookie and crumb from `fc.yahoo.com`	`short_interest` (fallback)	per symbol
yahoo_news	`query1.finance.yahoo.com/v1/finance/search`	none	`news`	—
yahoo_bars	chart API (1-minute, daily), with or without extended hours	none	bars	shallow 1-minute history (weeks); years of daily
alpaca_data	`data.alpaca.markets/v2/stocks/bars`, `/v1beta1/news`	key	bars (fallback), `news` (Benzinga)	IEX feed on the free plan: thin volume, about a week of intraday history, regular hours
alpaca_assets	assets API	key	`borrow` (shortable, easy to borrow)	**as of now only**: no history; a backtest applies today's flag and says so
alpaca_account	`GET /v2/account`, `GET /v2/account/activities/FILL` (paper or live base)	the account's own key, read-only	journal fills, key test	at most 50 pages × 100 fills per fetch today
finra	`cdn.finra.org/equity/regsho/monthly/shrt{date}.txt`, then the OTC biweekly paths	none	`short_interest` (shares short; % from the card's float)	**one file for every symbol**, fetched once per day; twice-monthly settlement
nasdaq_si	`api.nasdaq.com/api/quote/{t}/short-interest`	none	`short_interest` (fallback, two weeks of history)	per symbol
nasdaq_regsho	`nasdaqth{yyyymmdd}.txt`	none	`reg_sho`	per settlement day
nyse_regsho	NYSE threshold-securities download	none	`reg_sho`	per settlement day
edgar	`data.sec.gov/api/xbrl/companyfacts/CIK{…}.json`; `www.sec.gov/files/company_tickers.json`; submissions (SIC, 8-K items); `efts.sec.gov/LATEST/search-index`	a descriptive User-Agent (`EDGAR_UA`); 403 without it	`oneil_c`, `oneil_a`, `oneil_s` shares, `sic`, 8-K events, `news` (full-text search)	rate-limited; the five EDGAR traps (ONEIL.md §3.4); filings land weeks after quarter end
sec_13f	the SEC's quarterly Form 13F datasets (`SUBMISSION.tsv`, holdings)	User-Agent	`oneil_i`	quarterly, 45 days after quarter end; managers over $100M only
polygon	grouped daily (whole market); aggregates	key	`bars_universe`, bars	the free plan is end of day: no today's bars
finnhub	`finnhub.io/api/v1/company-news`	key	`news`	—
google_news	`news.google.com/rss/search`	none	`news`	asked only when the others return fewer than 2 items
openfda	drug approvals, device 510(k) / PMA	none	N events (planned)	—
usaspending	contract awards	none	N events (planned)	—

Adapter	Endpoint(s)	Auth	Families	Notes and quirks
broker_files	uploaded TradingView, TTP, DXtrade files	—	journal fills	one parser per format, each with a sample file (JOURNAL.md)

4. Bars: feeds, cache and limits

One loader shape for every consumer (chart, strategies, backtests, alerts, outcomes, the journal): a UTC time-indexed frame of open, high, low, close, volume.

Feed	Source	Strength	Weakness
`yahoo`	Yahoo	free, current, years of daily	shallow 1-minute history
`yahoo_ext`	Yahoo with pre- and after-market	one source, more hours	—
`polygon`	Polygon	deep history, extended hours, consolidated tape	free plan: no today's bars
`alpaca`	Alpaca IEX	live	thin IEX volume, short intraday history, regular hours
`hybrid`	Polygon history + Alpaca today	deep and today	**volume seam**: IEX volume is a fraction of Polygon's, which distorts every VWAP on the Alpaca part
`hybrid_yahoo`	Polygon history + Yahoo today	deep and today, both consolidated	—

- **The default feed is chosen by hand** and stored as a setting; without a choice it is `yahoo`. It used to be inferred from "a Polygon key exists", and a key turned out not to mean the plan covered the data: every 1-minute request was refused while the platform called Polygon its best feed. No route has its own default (Q-N6).
- **Warm-up**: the fetch window is extended so every indicator has enough history before the first bar shown. **As of a day** hides every bar after it.
- **The cache** (Parquet, per feed): a ceiling (default 512 MB, trimmed to 70 % when over) and a free-disk floor (default 1 GB, trimmed hard below it), swept after every write. A full disk once stopped every program on the box.
- **Outcome bars** for R3 are stored once per ticker per day and shared by every tool (REGISTERS.md §4.4).

5. Fallback orders

Family	Order	Rule
`quote`	TradingView	the batch is refused on misalignment; the last quote stays with its age
`short_interest`	FINRA → Yahoo → Nasdaq	stop at the first plausible answer; record `shortSrc`; `basis = outstanding` when no float (understates)
`borrow`	Alpaca	`unavailable` never means "not shortable"
`reg_sho`	Nasdaq + NYSE	both asked; either listing counts
`news`	TradingView, Alpaca, Yahoo, EDGAR, Finnhub in parallel; Google when fewer than 2 items	the failing ones are named; the others still count
outcome bars	Yahoo → Alpaca	per ticker
chart bars	the chosen feed	never silently replaced
index bars (M)	Yahoo	—
whole-market bars (RS)	Polygon	—

A second source for the same fact is **stored beside the first**, with the priority recorded, never overwriting it (the F-21 lesson).

6. Health: the data check made permanent

The legacy data check asked each source for data that **cannot** legitimately be empty and judged the content, when a button was pressed. In the new design the same checks run **on a schedule** (each adapter at its own cadence, and all of them before the open) and feed one health page:

Check	down	degraded
TradingView scanner: `close > 0`, 50 rows	zero rows; first row has no usable price; fetch failed	—
news: every source for a liquid ticker	no stories from any	some sources failed (with the key named)
Alpaca key: the account	rejected (401/403)	—
bar feeds: 40 daily bars each	fetch failed; no data; under 15 rows; a close NaN or ≤ 0; every volume zero; last bar over 5 sessions old; out of order or duplicated; a one-day move over 25 %	over 5 % of sessions without volume
indexes ^GSPC, ^IXIC	as bar feeds; “M cannot be built”	—
cross-source agreement (the only check that catches a confidently wrong feed)	worst close difference over 0.5 %	under 3 shared sessions; fewer than 2 working feeds (“configure a second feed”)
RS universe	no sessions; under 1,000 symbols	under 253 sessions
research facts	missing a required field	older than their clock allows, counted in trading sessions (M and groups: not from the last completed session; SIC: 14 days). The legacy check used 36 hours by the clock, which reads every weekend as stale
EDGAR	ticker map under 5,000; no quarters for a known filer; 403	—
sector map	—	a card sector with no ETF mapping (MARKET.md §4)

Each failing check carries its **fix**. Checks run **inside the services’ own environment**, so they never report a different set of keys from the one in use (the first live data check did). A red check sends an alert; amber shows on the page.

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7. Keys

Key	Used by
Alpaca (data)	`alpaca_data`, `alpaca_assets`
each journal account’s Alpaca key	`alpaca_account`, per account
Polygon	`polygon`
Finnhub	`finnhub`
EDGAR User-Agent	`edgar`, `sec_13f`

All in **one secret store** (a file of mode 0600, or a managed store), read by the services at start. Never in git, a log, a reply to a page or a backup. A missing key makes its adapter `unavailable` with the reason “key not set”, and the health page names it. The Settings page can set and test keys (the legacy page had no inputs: D-T4).

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8. What failed

ID	Fault	Prevented by
—	the same ticker fetched once per tool	facts fetched once, shared
D-S1	short interest and borrow wiped on a re-quote	facts kept apart from the quote
D-S2, F-14	a missing value read as a number or as bearish	the honest null
D-S3	a column shift misaligning every field	the alignment check (kept)
Q-N6	routes with their own Polygon default	no per-route defaults
—	“alpaca ok” printed daily while the key was refused	health judged by content (kept, and scheduled)
—	the data check ran only when pressed, and once from the wrong environment	scheduled, inside the services’ environment
D-T4	the fix lines point to a Settings section with no inputs	keys set and tested in Settings
—	a full disk from an unbounded cache	the cache limits (kept)

9. Contracts

Owns

Table	Key
`source_calls` (counters, last error)	adapter, day
`health_checks`	check, instant
`bars_cache` (on disk, Parquet)	feed, ticker, timeframe, range

Serves

Facts to the card engine, the market engine, the research worker, register jobs, quant and the journal, through the adapter interface: `fetch(family, subjects, as_of?)` returning facts with status, source and `known_at`.

Routes (desk-web)

Route	Returns
`GET /api/health`	every check’s latest result, with fixes
`POST /api/health/run`	run every check now
`GET /api/sources`	adapters, their calls, failures and last errors
`GET /api/cache`, `POST /api/cache/{sweep,clear}`	the bar cache

10. How the rebuild is proven

1. **Fixtures for every adapter**, including a failure, a timeout and a malformed answer, run in CI.
2. **Call count.** On a normal morning, the number of short-interest, borrow and news calls falls to one per ticker per family, against one per tool today (measured from the logs).
3. **Health parity.** The scheduled checks give the same verdicts as the legacy data check on the same morning.
4. **Missing key test.** Remove the Polygon key: its adapter shows `unavailable · key not set`, the RS step reports it, nothing crashes.